Physics-Derived Features Beyond Time Controls for Future-Time Prediction on Unseen Rooftop PV Systems

Chengyi Zhou

Abstract

Data-driven photovoltaic (PV) power models can appear accurate when training and test data share installations or timestamps, but new rooftop systems require transfer across both system and time. This study tests whether physics-derived representations add predictive value beyond measured weather and ordinary clock and calendar controls. The HKUST rooftop PV analysis retains 1,370,191 modeling-ready daylight observations from 37 SolarEdge systems. The primary experiment trains on pre-2023 observations from other systems and tests held-out systems during 2023, covering 612,371 test rows. Weather-only Model A obtains macro nRMSE of 0.1330; Model C, which adds cyclic hour-of-day and day-of-year features, reaches 0.1156; and physics-enhanced Model B reaches 0.1114. Ordinary time features account for 80.6% of the original Model A-to-B error gap. Model B provides a further 3.6% nRMSE reduction relative to Model C (paired-bootstrap 95% interval: 2.9%-4.4%), lowers nRMSE on 34 of 37 systems, and lowers MAE on all 37. Because model inputs are station-invariant at a shared timestamp, the design estimates a fleet-representative prediction curve rather than individualized roof behavior. A secondary same-period comparison is more accurate in absolute terms but allows exact timestamp-feature overlap. The results support a modest incremental benefit from the physics representation within one campus, conditional on contemporaneous measured weather; they do not establish individualized, day-ahead, or cross-climate forecasting.

Keywords: photovoltaics; physics-informed machine learning; XGBoost; cyclic time controls; domain generalization; cold-start prediction

1. Introduction

Rooftop photovoltaic generation is inherently variable because the amount of electricity produced at a given moment depends on solar geometry, atmospheric attenuation, clouds, temperature, wind, installation capacity, orientation, and equipment condition. This variability complicates monitoring and planning for distributed solar fleets. Prediction models can help estimate expected production, detect abnormal behavior, and support grid and building operations, but their usefulness depends on how they are evaluated. A model that performs well on randomly held-out timestamps from a familiar system may be learning that system's characteristic scale and response rather than a relationship that transfers to a new installation.

The cold-start setting is especially important for new or sparsely monitored rooftop systems. A newly commissioned system has no long history of measured output from which to learn site-specific behavior. In this study, however, cold-start has a bounded meaning: the model receives no target-system history and no system-specific inputs, so it estimates a fleet-representative normalized-power response to shared environmental conditions. It does not individualize predictions for roof orientation, shading, equipment, degradation, or maintenance. Prior research has shown that shared models can predict individual systems and that regional models can transfer across installations (Costa, 2022; Grzebyk et al., 2023). More recent work addresses zero-sample PV prediction with explicit domain-generalization architectures (Liu et al., 2025).

Physical knowledge offers one possible route to improved transfer. Measured global horizontal irradiance (GHI) captures current sunlight, but the same GHI value can occur under different solar positions and clear-sky expectations. Solar zenith and azimuth encode geometry; clear-sky GHI provides a time- and location-specific atmospheric reference; the clear-sky index expresses observed irradiance relative to that reference; and an estimated cell temperature represents the thermal conditions that influence PV conversion efficiency. These variables are derived from relationships that apply across systems and therefore may help a model rely less on installation-specific statistical patterns.

This paper evaluates that hypothesis with a controlled future-time feature comparison. Model A receives four measured weather variables. Model C adds ordinary cyclic hour-of-day and day-of-year encodings. Model B instead adds five physics-derived features: solar position, clear-sky context, and a thermal proxy. All three XGBoost regressors use the same target, grouped folds, algorithm, hyperparameters, and random seed. The primary Model B-versus-Model C difference therefore estimates the incremental predictive value of the physics representation beyond naive time information under the chosen protocol. XGBoost is appropriate because tree boosting captures nonlinear interactions in tabular data and has performed well in large PV fleets (Chen & Guestrin, 2016; Grzebyk et al., 2023).

The study makes four contributions. First, it implements system-grouped evaluation and a stricter protocol in which both held-out systems and target timestamps are absent from training. Second, it introduces an explicit weather-plus-time control so the physics representation is not credited for merely supplying clock and calendar information. Third, it documents that all model inputs are station-invariant at a shared timestamp, which prevents station fingerprinting but limits the task to a fleet-representative prediction curve. Fourth, it reports system-level macro metrics and paired uncertainty so installations with more observations do not dominate the result.

The primary future-time result shows that ordinary time controls reduce macro nRMSE from 0.1330 for weather-only Model A to 0.1156 for Model C, a 13.1% reduction. Physics-enhanced Model B reaches 0.1114, a further 3.6% reduction relative to Model C with a paired-bootstrap interval of 2.9%-4.4%. Model B improves nRMSE on 34 of 37 systems and MAE on all 37. The earlier same-period comparison remains a secondary benchmark: its 0.0919-to-0.0790 difference measures the value of the complete hybrid feature bundle under timestamp overlap, not an isolated physics effect.

2. Problem Definition and Hypotheses

The prediction unit is one daylight observation for one rooftop PV system. For system s at timestamp t, the response variable is the measured AC power P divided by the system's installed capacity C. Capacity normalization places systems of different sizes on a comparable scale:

$$y(s,t) = P(s,t) / C(s)$$

The normalized target is approximately interpretable as a fraction of installed capacity. It is not a capacity factor averaged over a long period; it is an instantaneous normalized power value. Rows outside the accepted range from -0.001 to 1.2 are marked impossible and excluded from modeling. The small negative tolerance accommodates numerical or measurement noise, while the upper bound allows modest output above nominal capacity without admitting extreme values.

The study uses two nested evaluation conditions. The primary future-time evaluation withholds complete systems and requires every test timestamp to occur after every training timestamp. For each fold, the model is fitted on modeling-ready rows through December 31, 2022 from the non-held-out systems and evaluated on 2023 rows from the held-out systems. The secondary same-period evaluation also withholds complete systems but allows training and test systems to share timestamps. This ordering distinguishes the main future-time claim from the easier same-period benchmark.

Three feature sets define the primary controlled comparison. Model A uses measured GHI, ambient temperature, wind speed, and relative humidity. Model C adds sine and cosine encodings of local hour of day and day of year. Model B uses the four weather variables plus apparent solar zenith, solar azimuth, clear-sky GHI, clear-sky index, and a GHI-based Faiman cell-temperature estimate. The null and alternative hypotheses are:

- H0: Physics-enhanced Model B does not reduce mean system-level nRMSE relative to weather-plus-time Model C in the future-time evaluation.
- H1: Physics-enhanced Model B reduces mean system-level nRMSE relative to weather-plus-time Model C in the future-time evaluation.
- H2: Any Model B advantage remains broad across held-out systems and folds when both systems and target timestamps are absent from training.

The primary outcome is system-level nRMSE averaged equally across held-out systems. Secondary outcomes are normalized MAE and MBE. Models are compared in paired form because they use the same systems, folds, timestamps, and eligible rows. The primary contrast is Model B versus Model C; Model C versus Model A quantifies the value of ordinary time context, and Model B versus Model A preserves the total feature-bundle comparison. Neither protocol forecasts future weather or evaluates operational day-ahead power.

3. Related Research

3.1 Data-driven PV prediction

Machine-learning models are attractive for PV prediction because the mapping from meteorological inputs to power is nonlinear and installation-dependent. XGBoost constructs an additive ensemble of decision trees and includes regularization and subsampling mechanisms that support scalable learning from large tabular datasets (Chen & Guestrin, 2016). In a study of 1,102 residential systems, Grzebyk et al. (2023) selected XGBoost for individual yield nowcasting because of its performance and practical simplicity. Their results demonstrated that one fleet-level model can outperform commercial software for individual systems, supporting the use of a shared learner rather than a separate model per installation.

Deep sequence models provide another path. Costa (2022) compared LSTM, convolutional, and hybrid convolutional-LSTM networks for household PV generation and evaluated whether models trained on regionally aggregated data could predict individual systems. The reported results supported regional-to-individual transfer. That study is relevant because it treats generalization as an explicit evaluation question, rather than assuming that accuracy on randomly held-out observations implies transfer.

3.2 Physics-based and physics-derived representations

A physics-based PV workflow typically transforms time and location into solar position, estimates clear-sky or actual irradiance components, estimates module temperature, and then maps irradiance and temperature to electrical output. The open-source pvlib library provides modular implementations of solar-position, irradiance, thermal, and electrical models, allowing researchers to use physical calculations either as complete models or as components of hybrid pipelines (Anderson et al., 2023). This study uses pvlib to construct physically meaningful inputs while leaving the final weather-to-power mapping to XGBoost.

The clear-sky reference uses the Ineichen-Perez formulation. Ineichen and Perez (2002) developed an airmass-independent formulation of the Linke turbidity coefficient and associated clear-sky irradiance models. Dividing observed GHI by modeled clear-sky GHI produces a clear-sky index that partially separates predictable solar geometry from cloud-related attenuation. This representation can make the meaning of a given irradiance value more comparable across hours and seasons.

Temperature is another important physical pathway because PV conversion efficiency changes with cell temperature. The Faiman model estimates module or cell temperature using incident irradiance, ambient temperature, wind speed, and empirical heat-loss coefficients (Faiman, 2008). The current implementation applies the default coefficients $u0 = 25.0$ and $u1 = 6.84$. However, it substitutes GHI for plane-of-array irradiance because the system-level records may aggregate arrays with multiple or uncertain orientations. This approximation creates a useful thermal proxy but should not be interpreted as a fully specified module temperature model.

3.3 Generalization to unseen PV systems

The literature increasingly distinguishes interpolation within familiar sites from prediction on unknown systems. Liu et al. (2025) proposed a generative adversarial domain-generalization network for zero-sample prediction and evaluated nine geographically and capacity-diverse PV systems. Their results show that unseen-system prediction is now an active research area rather than an entirely untested problem. The present study therefore does not claim to be the first cross-system PV model. Its contribution is a transparent controlled ablation on a larger set of installations within one campus, designed to quantify the incremental value of a compact physics feature set in a standard XGBoost pipeline.

This distinction matters for novelty. Complex domain-generalization models may maximize predictive accuracy, but they combine architecture, augmentation, and representation changes. A simpler paired comparison can answer a different question: how much improvement is obtained by adding five physically interpretable variables while holding the learner fixed? A positive answer would support physics-derived feature engineering as a low-complexity baseline for future domain-generalization research. A negative answer would be equally informative because it would show that physical transformations do not automatically improve transfer.

3.4 Research gap

Existing studies demonstrate fleet-level prediction, regional generalization, and advanced zero-sample domain generalization, but they do not determine the value of this exact physics feature set under a system-grouped ablation on the HKUST rooftop dataset. The research gap is therefore dataset- and design-specific: the incremental cross-system benefit of solar position, clear-sky context, and a thermal proxy has not been quantified for these campus rooftops under an otherwise identical XGBoost configuration. The study addresses that bounded gap and avoids a broader priority claim that the literature cannot support.

4. Dataset and Data Preparation

4.1 Source dataset and analytical subset

Lin et al. (2025) released a three-year rooftop PV dataset collected at the Hong Kong University of Science and Technology. The source contains inverter-level generation measurements from 60 gridconnected rooftop stations at five-minute resolution, on-site meteorological measurements at one-minute resolution, and system metadata. The campus is located in Hong Kong's humid subtropical climate. The source paper explicitly notes that the single-location climate may limit the generalizability of models trained on the data.

The current project uses 37 SolarEdge systems out of the 60 stations in the dataset and combines their records into a table with 2,756,516 rows aligned to 15-minute timestamps. This optimizer-equipped SolarEdge cohort provides a consistent hardware scope for the controlled cross-system analysis; the paper makes no claims about stations outside this cohort.

Table 1. Dataset scope after preprocessing

Quantity	Value	Interpretation
Source period	2021-2023	Three years of campus operation
Source stations	60	All rooftop stations described by Lin et al. (2025)
Analyzed systems	37	SolarEdge systems in the combined modeling table
Raw combined rows	2,756,516	Before daylight filtering
Daylight rows	1,387,682	Apparent zenith below 90 degrees
Modeling-ready rows	1,370,191	After exclusion-quality flags
Flagged daylight rows	17,491	Retained for audit but excluded from modeling

Source: Study analysis outputs; source-dataset description from Lin et al. (2025).

4.2 Timestamp handling and daylight filtering

Timestamps are localized to the Asia/Hong_Kong time zone rather than converted from another zone. Solar position is calculated for the campus coordinates 22.3364 degrees north, 114.2654 degrees east, at an assumed altitude of 60 m. Apparent solar zenith and azimuth are merged back to all system rows by timestamp. Only observations with apparent zenith below 90 degrees are retained. Removing nighttime rows avoids a misleading reduction in error from long sequences in which both observed and predicted power are trivially zero.

The daylight filter retained 1,387,682 of 2,756,516 rows. The maximum retained zenith was 89.9997 degrees. A single campus weather station supplies the measured weather, and the solar and clear-sky variables depend only on timestamp and campus location. The analysis verifies that model inputs do not vary among stations at a shared timestamp. Consequently, within a fitted fold, held-out systems observed at the same timestamp receive identical predictions; separately fitted folds can produce different prediction series. This structure prevents station fingerprinting through the inputs but leaves roof-specific variation in normalized output irreducible under the design.

4.3 Physics and time feature construction

Clear-sky GHI is computed for every retained timestamp with pvlib's Ineichen model. The clear-sky index is measured GHI divided by clear-sky GHI and is capped at 2.0. A GHI-based cell-temperature proxy uses the Faiman model with ambient temperature, wind speed, u0 = 25.0, and u1 = 6.84. Model C's ordinary time controls are sine and cosine of local fractional hour on a 24-hour cycle and sine and cosine of local day of year on a 365.2425-day cycle. These cyclic encodings preserve adjacency across midnight and the year boundary. Normalized power is measured power divided by system capacity. No target-system power values are used to construct weather, time, or physics features.

4.4 Quality control

The preprocessing pipeline is non-destructive: quality-control flags remain in the cleaned table, while the modeling-ready indicator determines which rows enter training and evaluation. A daytime fault zero is defined as zero power when GHI is at least 200 W/m2. A stuck measurement is an exactly repeated positive power or weather value lasting at least 16 consecutive 15-minute samples, equivalent to four hours. Additional rules flag missing model variables, nonpositive capacity, negative GHI or wind speed, relative humidity outside 0-100%, and normalized power outside -0.001 to 1.2.

The executed quality summary identified 16,991 rows with missing values and 500 daytime fault-zero rows. It detected no impossible values and no four-hour stuck power or weather runs. Seven hundred thirty-one rows occurred after a detected data gap, but this informational flag did not exclude rows because the tabular models do not use lagged features. In total, 17,491 daylight rows were excluded from modeling. The exact thresholds are transparent and reproducible, but they remain analyst choices; sensitivity to the fault-zero and stuck-sensor thresholds is a potential robustness check.

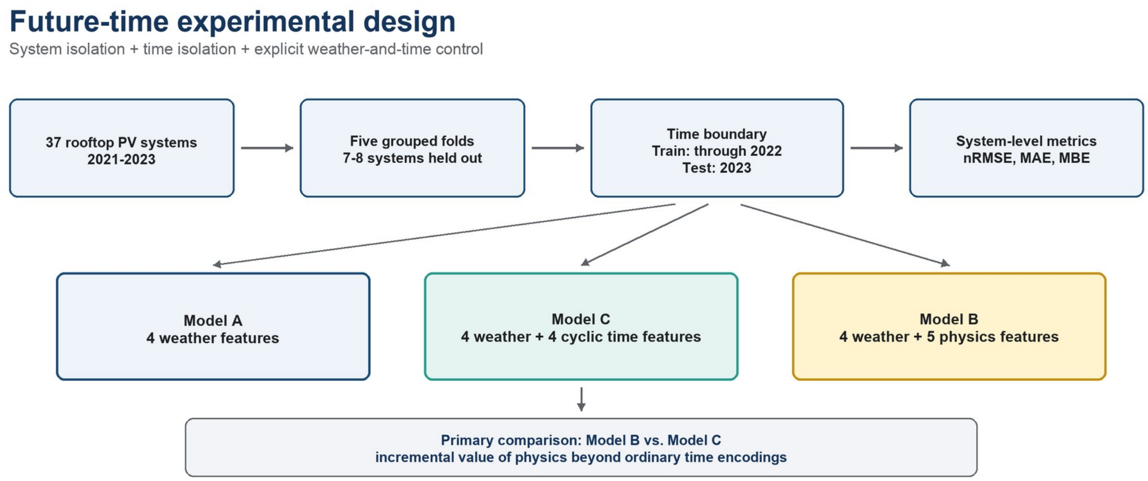

Figure 1. Primary future-time experimental design. Complete PV systems are held out, training ends in 2022, and testing uses 2023 observations. Models A, C, and B use identical folds and XGBoost settings. The primary Model B-versus-Model C comparison estimates the incremental value of the physics representation beyond ordinary time encodings. Source: Study design.

5. Methodology

5.1 Feature-ablation design

The principal experiment is a controlled future-time feature comparison. Models A, C, and B use the same regression algorithm, objective, hyperparameters, capacity-normalized target, model-ready rows, grouped folds, and random seed. Model A contains four measured weather features. Model C adds four cyclic time features, while Model B adds five physics-derived features. Plane-of-array irradiance is excluded because a system-level record may combine arrays with different or uncertain orientations. The same-period experiment retains the original Model A-versus-Model B bundle comparison as a secondary benchmark.

Table 2. Controlled feature sets for the primary future-time experiment

Feature	Model A	Model C	Model B	Rationale
Measured GHI	Yes	Yes	Yes	Current solar resource
Ambient temperature	Yes	Yes	Yes	Environmental and thermal context
Wind speed	Yes	Yes	Yes	Convective cooling context
Relative humidity	Yes	Yes	Yes	Atmospheric context
Hour sine/cosine	No	Yes	No	Cyclic time-of-day control
Day-of-year sine/cosine	No	Yes	No	Cyclic seasonal control

Feature	Model A	Model C	Model B	Rationale
Apparent zenith	No	No	Yes	Solar elevation geometry
Solar azimuth	No	No	Yes	Solar direction geometry
Clear-sky GHI	No	No	Yes	Time/location irradiance reference
Clear-sky index	No	No	Yes	Observed irradiance relative to clear sky
GHI-based cell temperature	No	No	Yes	Approximate thermal state

Source: Feature definitions used in the study analysis.

5.2 Reference baselines

The training-mean baseline predicts the mean normalized power of the training rows in each fold. It is intentionally weak but establishes the error of a constant predictor under the same split. The GHI physics baseline first computes a solar signal equal to clear-sky GHI divided by 1,000 and multiplied by the clear-sky index. Because the index reconstructs observed GHI up to the clipping rule, the signal is a capacity-normalized irradiance proxy. A nonnegative scalar is fit through the origin using only training rows, and predictions are clipped between 0 and 1.2. The fitted scalar ranges from 0.6254 to 0.6382 across the five folds.

This baseline is not a full PVWatts or system-specific physical model. It does not use system tilt, azimuth, module type, inverter efficiency, or loss parameters. Its purpose is to test how far a simple physical irradiance relationship can go without target-system calibration.

5.3 XGBoost specification

All XGBoost models use squared-error regression with the same configuration. Gradient-boosted trees sequentially correct residual error while regularization and subsampling reduce overfitting (Chen & Guestrin, 2016). Tree models do not require feature standardization, so scaling is disabled. No separate tuning is performed for Models A, C, and B; tuning independently would confound the feature comparison.

Table 3. Shared XGBoost configuration

Parameter	Value	Purpose
objective	reg:squarederror	Squared-error regression
n_estimators	1,000	Number of boosting rounds
learning_rate	0.05	Shrinkage per tree
max_depth	6	Maximum interaction depth
min_child_weight	10	Minimum child-node weight

Parameter	Value	Purpose
subsample	0.8	Row subsampling
colsample_bytree	0.8	Feature subsampling per tree
reg_alpha	0.0	L1 regularization
reg_lambda	1.0	L2 regularization
tree_method	hist	Histogram-based training
random_state	42	Deterministic split/model seed

Source: Model configuration used in the study analysis.

5.4 Leave-systems-out evaluation

Five-fold GroupKFold cross-validation is used with system ID as the group variable. The random seed is 42 and shuffling is enabled. Each fold contains approximately 29-30 training systems and 7-8 test systems. Every system appears in a test fold exactly once. Explicit assertions verify that the training-system and test-system sets are disjoint and that each expected system receives one evaluation result. This design follows the grouped-validation principle implemented in scikit-learn (Pedregosa et al., 2011).

Predictions are evaluated at fold and system levels. Headline values are macro means across the 37 system results, giving each system equal weight. Because no input identifies a roof, each fitted fold model emits a common prediction for systems sharing a timestamp. The estimand is therefore the accuracy of a fleet-representative environment-to-normalized-power relationship on an unseen roof, not the accuracy of an individualized roof model.

5.5 Performance metrics

Because the target is already divided by installed capacity, all errors are expressed as fractions of capacity. For observed normalized power y_i and prediction y-hat_i over N observations for a system, the metrics are:

$$nRMSE = \sqrt{[(1/N)\ \Sigma_i\ (\hat{y}_i - y_i)^2]}$$

$$MAE = (1/N)\ \Sigma_i\ |\hat{y}_i - y_i|$$

$$MBE = (1/N)\ \Sigma_i\ (\hat{y}_i - y_i)$$

Lower nRMSE and MAE indicate greater accuracy. MBE is signed: positive values indicate average overprediction and negative values indicate average underprediction. The primary relative reduction is 100 times the Model C minus Model B error divided by Model C error. Model C versus Model A measures the time-control gain; Model B versus Model A reports the total feature-bundle gain.

5.6 Leakage controls and reproducibility

The pipeline keeps every system in one fold and fits any learned preprocessing on training rows only. Models share identical group assignments, and model-ready rows contain no missing inputs or active exclusion flags. Station ID and system metadata are excluded, and all supplied features are station-invariant at a shared timestamp, preventing station fingerprinting through the input matrix. This does not make every form of leakage impossible: in the same-period benchmark, other systems contribute labels for the exact test timestamps and identical feature vectors. The future-time evaluation removes this information overlap by asserting disjoint systems, non-overlapping timestamps, and a strict train-before-test boundary.

Reproducibility is improved: the future-time notebook, Model C control, split audit, fold results, system results, summaries, and paired comparisons are preserved together in the project. Fixed seeds make the folds and bootstrap deterministic. Dependency versions remain unpinned, and a clean end-to-end rerun has not yet been documented; final submission should record the environment and dataset retrieval procedure.

5.7 Paired statistical comparison

Uncertainty is calculated with a deterministic paired bootstrap at the system level. Ten thousand resamples of the 37 paired system results are drawn with replacement using seed 42. The primary future-time contrast is Model B minus Model C; the analysis also reports Model C minus Model A and Model B minus Model A. Resampling timestamp rows independently would be inappropriate because rows within a system are temporally dependent. The system bootstrap is still conditional on an exchangeability approximation: systems share weather and, within a fold, a common prediction series. Its intervals should not be interpreted as 37 independent replications or as cross-location uncertainty.

5.8 Future-time prediction on unseen systems

The future-time evaluation assigns every system to one of five deterministic held-out folds. Training uses only modeling-ready observations through December 31, 2022 from the non-held-out systems, while testing uses January 1 through December 31, 2023 observations from the held-out systems. Each fold contains seven or eight test systems and 117,255 to 133,631 test rows. Because four systems begin in 2023, the actual number of systems contributing historical training rows ranges from 26 to 28. Across folds, every one of the 37 systems is tested exactly once, totaling 612,371 future observations.

The future-time suite contains five models: a training-mean baseline; a time/solar-only diagnostic using solar zenith, azimuth, and clear-sky GHI; four-feature weather-only Model A; eight-feature weather-plus-time Model C; and nine-feature physics-enhanced Model B. Model C uses the four weather variables plus hour and day-of-year sine/cosine pairs. The four XGBoost models use the same configuration, and all five models use identical station folds and eligible test rows.

For the primary Model B-versus-Model C nRMSE comparison, the system bootstrap reports both the absolute macro difference and the relative reduction. The same procedure is applied to Model C versus Model A and Model B versus Model A. Timestamp rows are not resampled as independent observations, and the shared-campus dependence noted in Section 5.7 remains a limitation.

6. Experimental Results

6.1 Primary future-time evaluation with an explicit time control

The station-and-time-disjoint experiment is materially harder than the same-period benchmark. Weather-only macro nRMSE rises from 0.0919 to 0.1330, while hybrid nRMSE rises from 0.0790 to 0.1114. The mechanism is direct: in the same-period design, the model can see the exact test-time feature vector during training, labeled with other systems' normalized output. This is not target-system leakage, but it gives the model contemporaneous fleet information and makes the task easier. The 2023 evaluation removes that timestamp overlap.

Table 4. Primary future-time performance across 37 held-out PV systems

Model	Macro nRMSE	SD	Macro MAE	Macro MBE
Training mean	0.2095	0.0165	0.1780	0.0133
Time/solar-only XGBoost	0.1670	0.0082	0.1193	0.0116
Model A: weather-only XGBoost	0.1330	0.0242	0.0986	0.0670
Model C: weather + time XGBoost	0.1156	0.0161	0.0838	0.0340
Model B: physics-enhanced XGBoost	0.1114	0.0176	0.0790	0.0336

Source: Model C future-time summary and system-level results. Training uses pre-2023 rows from other systems; testing uses 2023 rows from held-out systems.

The five-model result separates ordinary time context from the physics representation. Weather-only Model A obtains macro nRMSE of 0.1330. Model C adds only cyclic hour and day-of-year variables and reaches 0.1156, a 13.1% reduction with a bootstrap interval of 10.8%-15.2%. Model B reaches 0.1114. In absolute terms, Model C recovers 80.6% of the original Model A-to-Model B improvement, while the physics representation accounts for the remaining 19.4%. The time/solar-only diagnostic remains less accurate at 0.1670 because it omits measured weather.

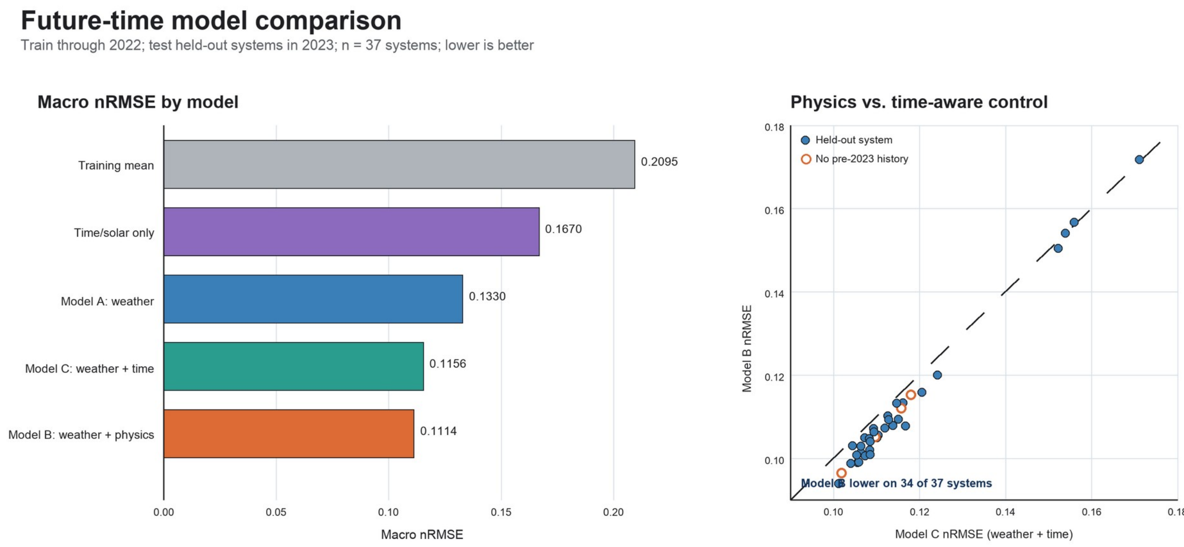

Figure 2. Primary future-time Model C comparison. The left panel shows macro nRMSE for all five models when training ends in 2022 and testing uses held-out systems in 2023. The right panel compares physics-enhanced Model B with weather-plus-time Model C; Model B is lower on 34 of 37 systems. Open circles identify four systems with no eligible pre-2023 history. Source: Model C future-time system-level results.

6.2 Physics adds a modest improvement beyond ordinary time controls

Physics-enhanced Model B reduces macro nRMSE from 0.1156 for weather-plus-time Model C to 0.1114. The absolute difference is -0.0042 of installed capacity, with a 95% interval from -0.0049 to -0.0035. The relative reduction is 3.6%, with a 95% interval of 2.9%-4.4%. Model B has lower nRMSE on 34 of 37 systems and in all five folds. Macro MAE falls from 0.0838 to 0.0790, a 5.7% reduction with an interval of 4.9%-6.6%, and Model B lowers MAE on all 37 systems. The original Model A-to-Model B nRMSE reduction remains 16.2%, but most of that total gain is attributable to the time context supplied by Model C.

6.3 No-history systems support the bounded cold-start result

The four systems with no eligible pre-2023 history also favor Model B over Model C. Relative nRMSE reductions are 3.2% for SQ Block P, 5.4% for Shaw Auditorium, 4.2% for UG Hall 8, and 2.3% for UG Hall 9. Their average nRMSE is 0.1113 for Model C and 0.1072 for Model B, a 3.7% reduction. This subgroup supports the bounded fleet-representative cold-start result, but four correlated campus systems do not establish individualized or cross-location transfer.

6.4 Bias remains unresolved beyond the time-aware control

Bias remains the main weakness. Weather-only Model A has macro MBE of 0.0670. Model C lowers it to 0.0340 and Model B to 0.0336. The Model B-minus-Model C bias interval spans -0.0010 to 0.0002, so the physics representation does not clearly improve mean bias beyond ordinary time controls. Most of the calibration improvement relative to Model A is therefore associated with time context, and both time-aware models retain positive 2023 bias.

6.5 Secondary same-period physics baseline

The calibrated GHI baseline reduced macro nRMSE from 0.2160 to 0.1237, a relative reduction of 42.7%. It reduced macro MAE from 0.1811 to 0.0776, or 57.2%. The GHI baseline had lower nRMSE than the training-mean predictor on all 37 held-out systems. This result is a useful validation of the experimental setup: a simple irradiance-based relationship contains transferable information that the constant baseline lacks.

The physics baseline nevertheless underpredicted on average, with macro MBE of -0.0170. It also remained materially less accurate than either XGBoost model. Therefore, physics alone provides a meaningful reference but does not capture all nonlinear relationships between weather and PV output in the analyzed systems.

6.6 Secondary same-period feature bundles reduce average XGBoost error

The secondary same-period analysis reports macro nRMSE of 0.0919 for weather-only Model A and 0.0790 for physics-enhanced Model B. The absolute difference is -0.0129 of installed capacity, a 14.0% relative reduction. Macro MAE falls from 0.0614 to 0.0522. Because Model A contains neither clock variables nor solar geometry, this comparison estimates the value of the complete weather-plus-physics-and-time bundle under timestamp overlap. It does not isolate a physics-specific effect.

Table 5. Secondary same-period performance across 37 held-out PV systems

Model	Macro nRMSE	SD	Macro MAE	Macro MBE
Training mean	0.2160	0.0185	0.1811	-0.0013
Calibrated GHI physics	0.1237	0.0146	0.0776	-0.0170
Model A: weather-only XGBoost	0.0919	0.0190	0.0614	0.0010
Model B: physics-enhanced XGBoost	0.0790	0.0228	0.0522	-0.0004

Source: Study analysis outputs.

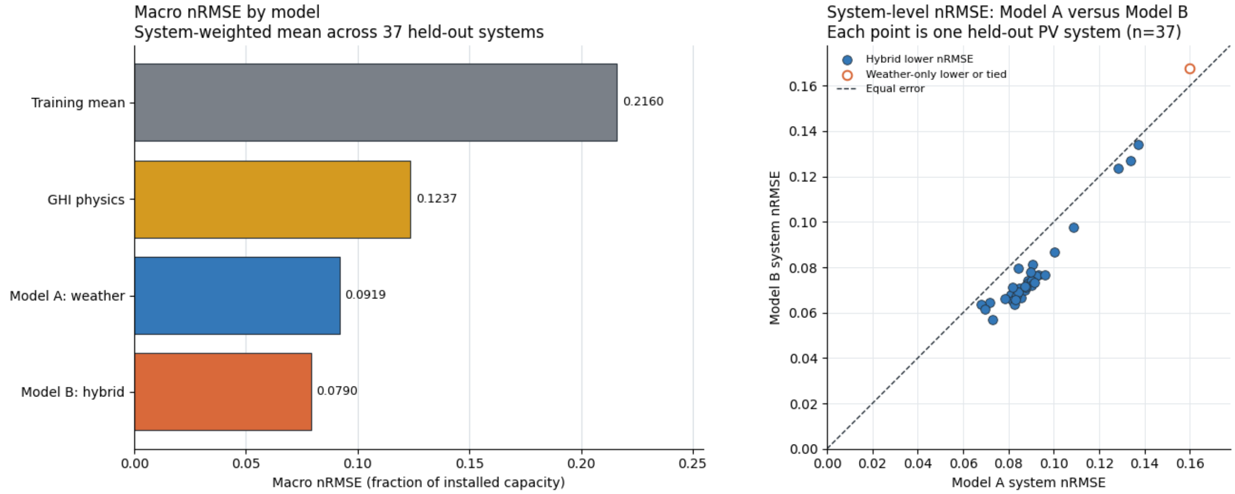

Figure 3. Secondary same-period feature-bundle comparison. The left panel shows macro nRMSE across four models. The right panel compares weather-only Model A with physics-enhanced Model B for each held-out system; 36 of 37 points favor Model B. Because training systems supply labels for the same timestamps, the figure does not separate physics from ordinary time context. Source: Study analysis outputs.

6.7 Same-period gains are broad but not universal across systems

Aggregate improvement is not the only relevant same-period result. Model B's system-level nRMSE standard deviation is 0.0228, compared with 0.0190 for Model A, and its maximum system error is slightly worse. The hybrid bundle lowers the macro average and median but does not improve every system. These results describe the bundle comparison under overlapping timestamps rather than an isolated physics mechanism.

This heterogeneity is scientifically important because reliability on a new system matters in addition to average accuracy. Model B has lower same-period nRMSE on 36 of 37 systems, but the systems share weather and feature vectors and should not be treated as independent confirmations. The scatterplot shows that the bundle gain is broad within the campus sample while one system favors Model A.

6.8 Same-period paired uncertainty supports the within-campus comparison

The same-period 14.0% reduction has a paired-bootstrap interval of 11.3%-16.5%, and the absolute Model B-minus-Model A interval is -0.0146 to -0.0109. These intervals favor Model B under the system-resampling procedure, but the shared timestamp inputs and weather weaken the independence approximation. The result is a descriptive secondary benchmark, not the paper's primary physics-beyond-time test.

7. Discussion

7.1 What the physics representation adds beyond time controls

The remaining Model B advantage is consistent with physical transformations providing context that naive time variables do not. Raw GHI does not state whether irradiance is high or low relative to the clear-sky expectation. Clear-sky GHI and clear-sky index add that reference; solar zenith and azimuth represent geometry more directly than hour and day encodings; and the cell-temperature proxy combines irradiance, temperature, and wind in a conversion-relevant form.

Model C materially changes the interpretation. Ordinary time features explain about 80.6% of the original weather-to-hybrid nRMSE gap. Model B nevertheless improves a further 3.6% relative to Model C, with a system-bootstrap interval excluding zero and lower nRMSE on 34 of 37 systems. The evidence therefore supports a modest incremental benefit from the complete physics bundle, not the claim that physics alone caused the full 16.2% reduction. Additional grouped ablations are needed to identify which physics component drives the residual gain.

7.2 Comparison with related work

The results align with prior evidence that shared models can predict individual PV systems (Costa, 2022; Grzebyk et al., 2023) and complement domain-generalization work by Liu et al. (2025). The present study uses a standard learner and 37 systems from one campus to compare representations transparently. Model C strengthens that comparison by distinguishing naive temporal context from the full physics feature bundle, although the datasets and prediction tasks remain different from prior studies.

The GHI baseline confirms that a simple irradiance-to-power relationship transfers better than a constant predictor. Weather Model A improves further, Model C adds substantial ordinary time context, and Model B provides a smaller additional gain. This ordering suggests that measured weather, predictable temporal structure, and physical transformations each contribute under the future-time protocol.

7.3 Practical meaning of the reported error

In the primary future-time experiment, Model B's nRMSE of 0.1114 means root-mean-square error of approximately 11.14% of installed capacity for the average held-out system. For a 100 kW system, that corresponds to roughly 11.1 kW RMSE under eligible daylight conditions. Model C's 0.1156 corresponds to about 11.6 kW, so the incremental physics benefit is approximately 0.42 kW RMSE for a 100 kW system under this normalization. Model B's MAE is about 7.9 kW and its MBE indicates average overprediction of 3.36 kW.

That interpretation should not be confused with day-ahead operational forecasting. The future-time evaluation predicts later timestamps, but the models receive measured contemporaneous irradiance and weather at those timestamps. Replacing those inputs with weather forecasts would introduce additional error and require a defined forecast horizon. The present model is most directly relevant to conditional prediction, performance benchmarking, and fault-screening applications.

7.4 Same-period evaluation is optimistic but directionally reliable

The same-period evaluation asks whether a feature bundle transfers across systems when other systems supply observations from the same dates. At a shared timestamp, the feature vector is identical across systems, so the model has already seen that exact test-time input paired with other roofs' output. This mechanism explains why same-period errors are lower. The benchmark remains useful for historical fleet estimation, but it cannot isolate physics from time and should not be the primary evidence for future-year transfer.

7.5 Why future-time error and bias remain system-dependent

Model B improves nRMSE over Model C on 34 of 37 systems, but absolute error remains heterogeneous. Zone A2, UG Hall 2 2F, and S H Ho Sports Hall slightly favor Model C in nRMSE, while Model B lowers MAE on all systems. Because the models receive no roof-specific features, common clear-sky and temperature proxies cannot represent orientation, shading, module technology, inverter behavior, degradation, or maintenance.

System-level diagnostics should compare error and bias with available metadata such as capacity, orientation complexity, commissioning date, and missing-data rate. Such variables are absent from the current predictors, so between-roof response differences remain unexplained. The near-equal Model B and Model C bias also suggests that calibration or transferable system metadata, rather than additional common timestamp features alone, may be needed to reduce the positive 2023 shift.

8. Limitations, Uncertainty, and Threats to Validity

Several limitations bound the conclusions. First, all analyzed systems share one campus, climate, weather station, and location-derived solar variables. Within each fitted fold, systems at a shared timestamp receive identical predictions. This prevents station fingerprinting and creates a clean fleet-level transfer test, but it does not evaluate individualized prediction or geographic, meteorological, and sensor-domain shift. Between-system variation in normalized output is irreducible without transferable system-specific inputs.

Second, both evaluations are conditional power prediction rather than complete operational forecasts. The future-time evaluation moves the target timestamps into a later year, but contemporaneous measured weather is still supplied to the models. Numerical weather-prediction inputs would add forecast error and require an explicit horizon. Future-time conditional prediction is therefore more accurate terminology than day-ahead forecasting.

Third, physical feature accuracy is constrained by metadata. Plane-of-array irradiance is omitted because the site-level power series can aggregate arrays with uncertain or mixed orientations. The Faiman temperature proxy therefore uses GHI, and its default coefficients were originally derived for open-rack modules under different conditions (Faiman, 2008). These choices favor robustness and availability over system-specific physical fidelity.

Fourth, the 15-minute rows within a system are temporally dependent, and systems share the same weather and, within a fold, the same prediction series. The 37 system metrics are therefore correlated targets, not 37 independent confirmations. System-level macro evaluation and paired system resampling use the correct primary unit, but the bootstrap's independence approximation is weaker than it would be across locations and may understate broader uncertainty. A time-block or two-way system-and-time bootstrap is an appropriate future sensitivity analysis.

Fifth, temporal robustness is evaluated with one cutoff and one future year. Four systems have no eligible pre-2023 history and provide direct evidence for the bounded no-history setting, but their predictions still follow the same fleet-representative timestamp relationship. Rolling cutoffs and additional years are needed to estimate year-to-year variability.

Sixth, the reported population is restricted to modeling-ready daylight observations. Quality-control rules exclude missing values, physically impossible ranges, stuck measurements, and likely daytime fault zeros; some exclusions therefore use observed power. The reported errors describe expected production during retained operation and should not be interpreted as performance for fault or outage detection.

Seventh, the future-time notebook, Model C result tables, and figures are archived with the experiment, but package versions remain unpinned and a clean end-to-end rerun has not been documented. The saved outputs are internally consistent and reproducible from the station-level CSV; final submission should record the environment and dataset retrieval procedure.

Finally, the shared XGBoost configuration was selected before the paired comparisons but not through nested tuning. Holding it fixed is appropriate for comparing Models A, C, and B, yet it does not show that any feature set is optimally tuned. The findings apply to one reasonable common configuration.

9. Future Work

The first priority is to move from contemporaneous measured weather to operational weather forecasts and to repeat the future-time experiment with rolling temporal cutoffs. A block-bootstrap sensitivity analysis over systems and days or weeks would also measure uncertainty under shared weather more conservatively.

A second priority is robustness across hardware and location. The evaluation should be repeated in an independent climate and with transferable system metadata such as orientation, shading, module type, inverter characteristics, and commissioning date. This would test both cross-location generalization and individualized cold-start prediction.

A third direction is mechanism-focused ablation. Adding solar geometry, clear-sky normalization, and temperature features in separate groups would identify which component produces the 3.6% incremental gain beyond Model C. Calibration should also be evaluated using training-system data only to address the positive 2023 bias without target-system history.

Finally, physics-as-features can be compared with physics-guided residual correction. In the latter design, a physical model produces an initial prediction and machine learning models its residual error. Comparing these strategies under the same station-and-time-disjoint folds would test whether physical structure is more transferable as an input representation or as a baseline model.

10. Overall Conclusion

This study asks whether physics-derived variables improve future-time PV power prediction beyond measured weather and ordinary time controls for systems absent from training. The secondary same-period benchmark gives macro nRMSE of 0.0919 for weather-only Model A and 0.0790 for Model B, but exact timestamp-feature overlap makes that comparison easier and prevents physics-specific attribution.

In the primary future-time evaluation, weather-only Model A obtains macro nRMSE of 0.1330, weather-plus-time Model C obtains 0.1156, and physics-enhanced Model B obtains 0.1114. Model C explains 80.6% of the original Model A-to-Model B gap. Model B provides a further 3.6% reduction relative to Model C (95% interval: 2.9%-4.4%), lowers nRMSE on 34 of 37 systems and all five folds, and lowers MAE by 5.7% on all 37 systems. Bias is not clearly better than Model C.

The most defensible conclusion is bounded: the complete physics representation provides a modest incremental accuracy benefit beyond naive time encoding in this campus dataset and controlled XGBoost pipeline. Most of the original weather-to-hybrid gain comes from temporal context. Because inputs are station-invariant, the model estimates a fleet-representative curve rather than individualized roof behavior; because contemporaneous weather is supplied and all systems share one climate, the study does not establish day-ahead or cross-climate forecasting. The shared prediction environment also means that 37 system-level results should not be interpreted as independent replications.

References

Anderson, K. S., Hansen, C. W., Holmgren, W. F., Jensen, A. R., Mikofski, M. A., & Driesse, A. (2023). pvlib python: 2023 project update. Journal of Open Source Software, 8(92), 5994. https://doi.org/10.21105/joss.05994

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (pp. 785-794). Association for Computing Machinery. https://doi.org/10.1145/2939672.2939785

Costa, R. L. de C. (2022). Convolutional-LSTM networks and generalization in forecasting of household photovoltaic generation. Engineering Applications of Artificial Intelligence, 116, 105458. https://doi.org/10.1016/j.engappai.2022.105458

Faiman, D. (2008). Assessing the outdoor operating temperature of photovoltaic modules. Progress in Photovoltaics: Research and Applications, 16(4), 307-315. https://doi.org/10.1002/pip.813

Grzebyk, D., Alcañiz, A., Donker, J., Zeman, M., Ziar, H., & Isabella, O. (2023). Individual yield nowcasting for residential PV systems. Solar Energy, 251, 325-336. https://doi.org/10.1016/j.solener.2023.01.036

Ineichen, P., & Perez, R. (2002). A new airmass independent formulation for the Linke turbidity coefficient. Solar Energy, 73(3), 151-157. https://doi.org/10.1016/S0038-092X(02)00045-2

Lin, Z., Zhou, Q., Wang, Z., Wang, C., Bookhart, D. B., & Leung-Shea, M. (2025). A high-resolution three-year dataset supporting rooftop photovoltaics (PV) generation analytics. Scientific Data, 12, Article 63. https://doi.org/10.1038/s41597-025-04397-y

Liu, S., Qi, Y., Li, D., Liu, L., Wang, S., Fernandez, C., & Gao, X. (2025). Adversarial multi-source domain generalization approach for power prediction in unknown photovoltaic systems. Applied Soft Computing, 181, 113495. https://doi.org/10.1016/j.asoc.2025.113495

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., &

Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825-2830. https://www.jmlr.org/papers/v12/pedregosa11a.html